# Pitching at Altitude: Which Pitchers Hold Up at Coors Field?

Elliot Malsam · 2025 Statcast analysis, tested on 2026 · elliotma27.github.io · github.com/elliotma27/coors-pitching-at-altitude

**In short:** Coors adds far more damage on fly balls than on ground balls, and fly-ball pitchers lose the most there, in both 2025 and 2026. Splitters, changeups, and sinkers keep the ball on the ground. My one-season pitcher projections did not predict the following season, which points toward scouting and developing stable traits like fly-ball rate rather than chasing single-season Coors splits.

## Question

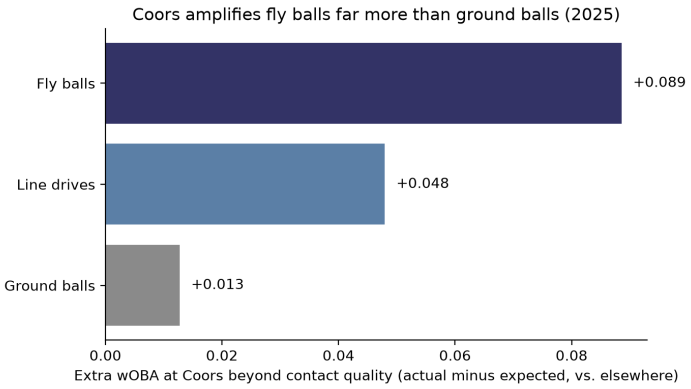
Which pitcher profiles lose the least at Coors Field, why, and which pitchers fit that profile?

## Data and method

2025 regular-season Statcast data (about 124,000 balls in play), pulled with pybaseball and analyzed in SQL (SQLite) and Python. xwOBA is built only from exit velocity and launch angle, so it does not know about altitude. The gap between actual and expected wOBA at Coors, compared with other parks, isolates where the park itself adds damage. I then grouped 311 pitchers (150+ balls in play) by fly-ball rate and hard-hit rate and compared their results at Coors versus elsewhere.

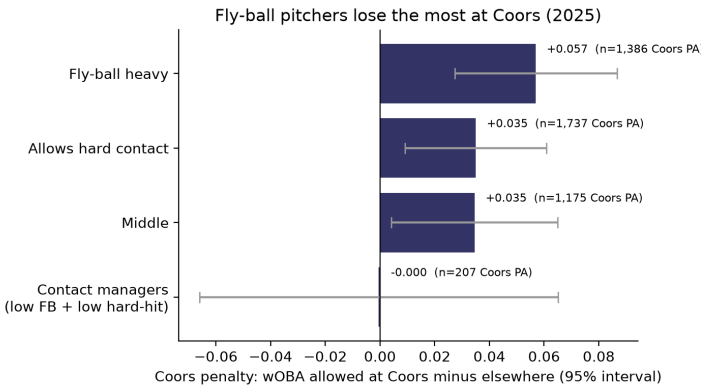
## Finding 1: Coors amplifies fly balls far more than ground balls

Relative to other parks, fly balls at Coors produced **+0.089 wOBA** beyond their contact quality, line drives +0.048, and ground balls only +0.013. Hitters were not making better contact at Coors (fly-ball xwOBA was .424 there vs. .430 elsewhere); the same contact simply did more damage.




## Finding 2: Fly-ball pitchers lose the most, and fly balls matter more than hard contact

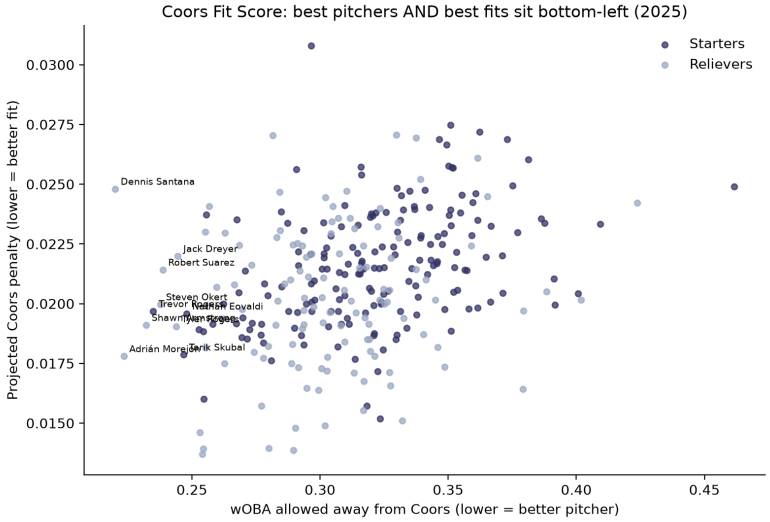
Fly-ball-heavy pitchers allowed **+0.057 wOBA** more at Coors (95% interval about ±0.030). Pitchers who allowed hard contact *without* a fly-ball tendency lost about the same as average pitchers (+0.035), so avoiding fly balls is what protects a pitcher at Coors. Contact managers (low fly-ball and low hard-hit rates) showed no penalty, but with only 207 Coors plate appearances that estimate is very uncertain. The profile is also rare: just 33 of 311 pitchers.



## Finding 3: A Coors Fit Score for every pitcher

To get around small Coors samples, I built a translation table of how much each exit velocity and launch angle combination gains at Coors (shrunk toward zero where Coors data is thin), then ran every pitcher's *road* batted balls through it. The largest boosts come from hard-hit balls at fly-ball angles (95 to 110 mph, 20 to 40 degrees). This scores all 296 qualified pitchers on a continuous scale, which surfaces near-fits instead of a simple yes or no.

**The best fits combine strikeouts and ground balls**, since a strikeout cannot carry in thin air. But projected penalties span a narrow range (+0.014 to +0.025 wOBA), while talent varies far more (road wOBA .22 to .38). **Fit is a tiebreaker between similarly talented pitchers, not the main driver.**




| Profile | Pitchers | Why they stand out |
|---|---|---|
| Coors-proof starters | José Soriano, Tylor Megill, Yoshinobu Yamamoto | Lowest projected penalties among starters |
| Best projected starters | Trevor Rogers, Tarik Skubal, Nathan Eovaldi | Elite talent plus a solid fit |
| Coors-proof relievers | Jhoan Durán, Abner Uribe, Brad Keller | Lowest projected penalties, strong road results |

*Full ranked list of all 296 pitchers in the GitHub repo (coors_fit_scores.csv).*

## Validation

Within 2025, projected and actual Coors penalties were essentially uncorrelated at the pitcher level (r = -0.03, 61 pitchers with 20+ Coors PA), because one season gives most pitchers only 20 to 40 batters at Coors. I then tested it out of sample: pitchers' 2025 scores did not predict their 2026 Coors results (see page 4). The honest conclusion is that the score describes contact profiles well but is not yet a reliable predictor for individual pitchers.

## What this makes me want to explore: scouting and development

If the profile that fits Coors is rare and expensive to acquire, the Rockies may get more value building it than buying it. Two ideas I would want to test:

- **Amateur scouting:** identify college and high school pitchers who are already excellent at limiting fly balls, using batted-ball data from college TrackMan systems where available, and use that as a tiebreaker between similar talents.
- **Development:** emphasize pitch traits tied to ground balls, such as sinkers, splitters, and changeups (see Study 2), for pitchers in the Rockies system, so they reach Denver with a Coors-ready profile.

Encouragingly, fly-ball rate is a fairly stable trait at the MLB level (r = 0.64 from 2025 to 2026), more stable than hard-hit rate (r = 0.50). The next question to answer: do fly-ball tendencies in college and the minors carry over to the majors?

## Limitations

Single-season Coors samples are small. Coors plate appearances skew toward Rockies pitchers, and visiting pitchers at Coors face the Rockies' own lineup, which was weak in 2025 and likely hides some of the park effect in actual results. Weather, the humidor, and reduced breaking-ball movement at altitude are not modeled. Archetype cutoffs are simple thirds.

## Next steps

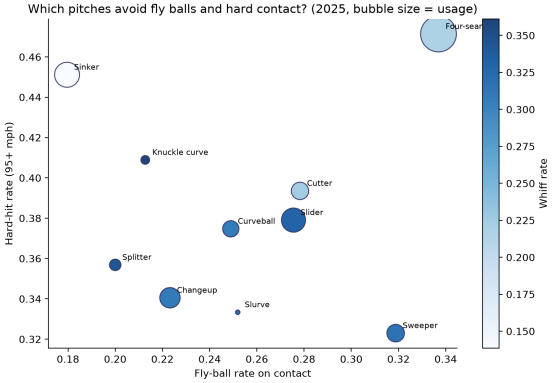
Pool more seasons, adjust for opponent quality, and add pitch-level traits like sinker usage and vertical break. The 2026 out-of-sample test is on page 4.

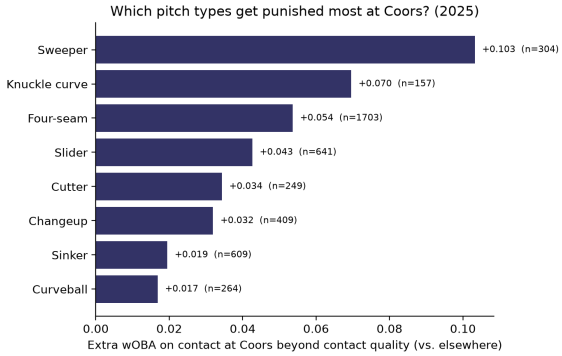
# Study 2: Which pitches protect against Coors, and does the Rockies' mix use them?

Same 2025 Statcast data. Pitch-type profiles computed league-wide in SQL; Coors effect measured as actual minus expected wOBA on contact at Coors vs. elsewhere.

## Finding 4: Splitters, changeups, and sinkers keep the ball on the ground; four-seamers get punished

Splitters combine a high whiff rate (.345) with a .531 ground-ball rate and low hard contact (.357), and lost almost nothing at Coors (+0.005, small sample). Changeups allowed the least hard contact of any major pitch (.341). Sinkers produced the most ground balls (.548) and one of the smallest Coors penalties (+0.019). Four-seamers allowed the most fly balls (.337), hard contact (.471), and barrels (.110), with a clear Coors penalty (+0.054 on 1,703 balls in play). The four-seam penalty repeated in 2026 (+0.039 on 1,449 balls in play). Sweepers were the most punished pitch at Coors in 2025 (+0.103) but not in 2026 (+0.020), so I treat that result as noise.



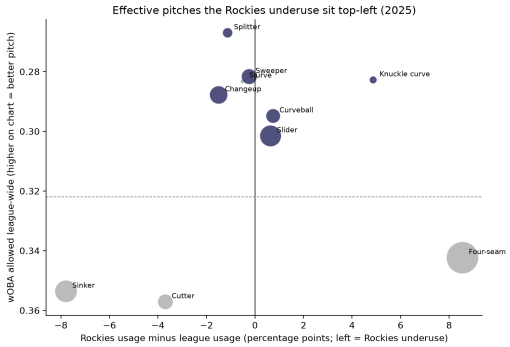




## Finding 5: In 2025, the Rockies leaned on the pitch Coors punishes most

Rockies pitchers threw four-seamers **40.6%** of the time vs. 32.0% league-wide (+8.6 points, the largest gap of any pitch), and sinkers at **half the league rate** (7.8% vs. 15.6%). Their mix was nearly identical at home and on the road (four-seamers 40.9% at home, 41.4% away), so this is not a deliberate altitude adjustment. Sinkers look average league-wide (.354 wOBA allowed) because they rarely miss bats, but at Coors, where keeping the ball on the ground matters most, they are more valuable than their league numbers suggest.

Two pitches are both underused by the Rockies and better than average league-wide: **changeups** (-1.5 points, .288 wOBA allowed) and **splitters** (-1.1 points, .267, the best results of any pitch type).

**2026 update:** the Rockies moved sharply toward the league mix. Four-seamers fell from 40.6% to 30.6% and sinkers nearly doubled (7.8% to 14.6%). One gap remains: they threw **fewer sinkers at home (12.6%) than on the road (16.4%)**, the opposite of what Coors would suggest.




## What this suggests for development

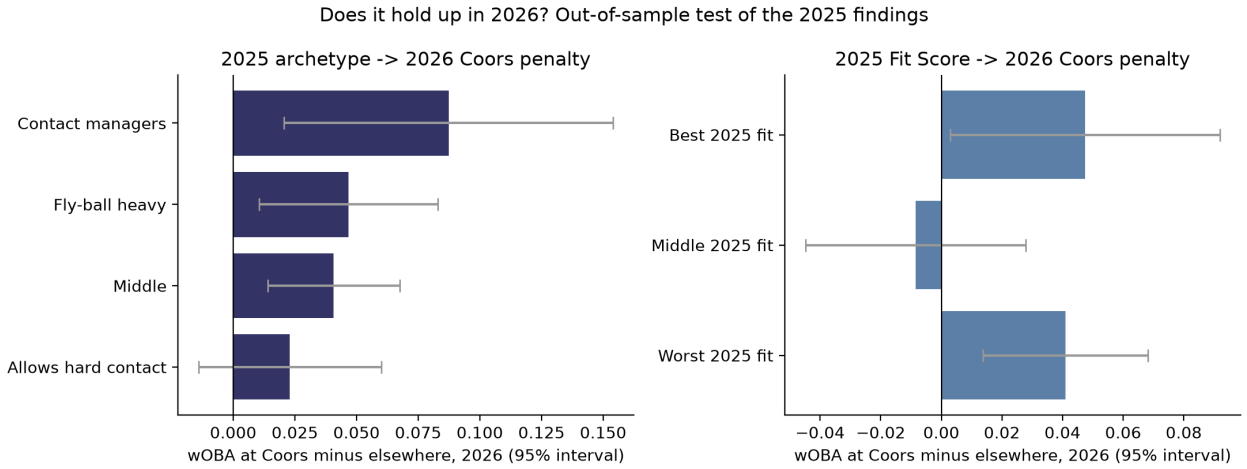
For pitchers headed to Coors, prioritize developing splitters, changeups, and sinkers, and where a pitcher's arsenal allows, rely less on four-seamers at home, and keep sinkers in the mix at Coors.

**Caveats:** pitch mix is limited by each pitcher's arsenal, so this is a development direction, not a usage mandate. League results partly reflect who throws each pitch, high-ride four-seamers likely fare better, the splitter's Coors estimate rests on 126 balls in play, and team assignment is approximate (28 Rockies pitchers in 2025, 34 in 2026).

# Does it hold up in 2026? What replicated and what didn't

I reran every step on the 2026 season, then tested whether 2025 labels and scores predicted 2026 results.

| Finding | 2025 | 2026 | Held up? |
|---|---|---|---|
| Coors boost on fly balls vs. ground balls | +0.089 vs. +0.013 | +0.056 vs. +0.018 | Yes |
| Fly-ball-heavy pitchers' Coors penalty (same season) | +0.057 (±0.030) | +0.047 (±0.023) | Yes |
| Contact managers' Coors penalty (same season) | 0.000 (±0.066) | +0.018 (±0.059) | Directionally |
| Four-seam Coors penalty | +0.054 | +0.039 | Yes |
| Sinker Coors penalty | +0.019 | +0.024 | Yes |
| Sweeper Coors penalty | +0.103 | +0.020 | No |
| 2025 archetype predicts 2026 Coors results | | Contact managers +0.087 (±0.067) | No |
| 2025 Fit Score predicts 2026 Coors results | | No pattern (r = -0.10) | No |
| Fly-ball rate stable year to year | | r = 0.64 (n = 190) | Yes |




## What I take from this

**The park effect and the trait are real; the individual predictions are not yet.** Across both seasons, Coors adds the most damage on fly balls, and within each season fly-ball pitchers lose the most. Fly-ball rate is also a fairly stable skill. But labeling a pitcher from one season and predicting his next season at Coors did not work: the 2025 contact managers actually did worst at Coors in 2026, on a small sample of 247 plate appearances, and the Fit Score showed no predictive pattern.

Two likely reasons: individual Coors samples are tiny (most pitchers face 20 to 40 batters there per season), and the contact-manager label requires two traits at once, so it reshuffles a lot from year to year even though fly-ball rate on its own holds steady.

**What I would do next:** pool three or more seasons, project pitchers from stable traits like fly-ball rate rather than one-season Coors splits, adjust for opponent quality, and test the score on the 2027 season before relying on it for individual decisions.